

TASK 8

Date:- 30/9/2025

Normalizing database using functional dependencies upto BCNF

Aim: Normalizing database using functional dependencies upto BCNF

mobile phone database:

1. Identify mobile phone attributes: mobile name, model name from mobile phone
2. Define relational schema: mobile phone (mobile name phone, id, phone no, model name from mobile phone)
3. Determine functional dependencies (FDs) between attributes:
IMEI no $\rightarrow$ model no, brand name, price, ram, storage, processor, release year
model no $\rightarrow$ Brand name, price, ram, storage, processor, release year
processor $\rightarrow$ Brand name

Step 2 :- Convert to 1NF

Rule :- Make sure each column has only one value per

$\rightarrow$ we separate the repeating groups into multiple rows

mobile phone (IMEI no, model no, Brand, Name, price, RAM, storage, processor, release year).

Step 3: Convert to 2NF

Rule:

A relation R is in 2NF if:

- $\rightarrow$ it is already in 1NF
- $\rightarrow$ no partial dependency exists (i.e. no nonkey attribute depends on part of a composite

Attribute	Description
model-no	model-name
brand-name	brand
price	model price
RAM	memory
storage	internal storage
processor	cpu
Release-Year	launch year

Step 4: Convert to 2NF

Rule: If $P \rightarrow N$ in 2NF

* There are no transitive dependencies $P \rightarrow N$
 no non prime attribute depends on the other non-prime attribute.

Relation	Attribute
mobile-phone	IMEI - no, model-no
model-detail	model-no, brand-name, price, RAM & storage
Brand	brand-name, processor

Step 5: Convert to 3NF

Rule:

Relation in 3NF (Boyce-Codd normal form)

— for every non-trivial functional dependency

$X \rightarrow Y$

X must be a Super key

Normalized Schema:

mobile-phone (IMEI-no, primary key, model-no
 — foreign key)

model - Details (model - no primary key, board name)

Board (Board - Name primary key, professor)

VEL TECH	
EX NO.	8
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	4
RECORD (5)	
TOTAL (20)	14
SIGN WITH DATE	

Result:

Thus, the normalizing database using functional dependencies upto BCNF has been completed successfully.